INKLOOM AI HACKATHON | ROUND 2: THE BUILD | WORKING MVP SUBMISSION

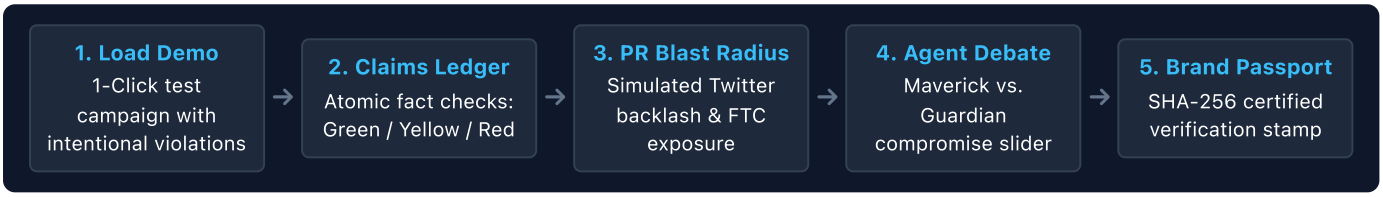
# BrandCircuit

The Multimodal Pre-Flight CI/CD & Verified Claims Ledger for Enterprise Brand Content

| ROUND | CORE WORKFLOW | REPOSITORY | EVALUATION WINDOW |
|---|---|---|---|
| Round 2 — The Build | 100% Functional & Testable | Git Stamped & Documented | 22–25 October 2026 |

## 1 Executive Summary & Core Working Workflow

**BrandCircuit** transforms enterprise brand governance from a slow, bureaucratic gatekeeper into an intelligent pre-flight continuous compliance engine. The Round 2 MVP provides a complete, end-to-end working experience accessible to judges with zero installation or configuration required.




## 2 List of Completed Features (Round 2 MVP)

| Feature Module | What is Built & Working | User Experience |
|---|---|---|
| **Verified Claims Ledger (SSOT)** | Extracts factual assertions, statistics, and guarantees. Validates against Single Source of Truth registry into 🟢 Verified, 🟡 Needs Citation, 🔴 Disputed. | Inline color-coded tags with direct 1-click "Resolve" shortcuts. |
| **PR Blast Radius & Crisis Simulator** | Simulates potential social media backlash (Twitter/X community note risk) and calculates projected regulatory penalty exposure (FTC §5). | Live crisis gauge & simulated public backlash thread. |
| **Maverick vs. Guardian Agent Debate** | Interactive multi-agent negotiation between Growth Marketing (high conversion) and Compliance Counsel (zero legal liability) with dynamic slider. | Slider dynamically updates copy across 3 levels of risk/virality. |
| **Multimodal Visual DNA Inspector** | Canvas overlay analyzing marketing banners for logo exclusion zone clear-space, color drift tolerance ($\Delta E$), and WCAG contrast. | Interactive modal showing real-time pass/fail metrics. |
| **Cryptographic Brand Passport** | Mints tamper-proof SHA-256 cryptographic verification tokens once content meets 100% compliance criteria. | Audit-proof certificate ready for CMS and social API dispatch. |
| **Role-Based Collaboration** | Instant toggle between Creator Drafting Mode and Reviewer Sign-Off Queue. | Header role switcher updating permissions in real time. |

## 3 Technology & Implementation Details

### 🖥️ Frontend Architecture

- **Responsive UI:** Tailwind CSS with dark-mode executive styling.
- **Interactive Workspaces:** Real-time reactive state management for the compromise slider and modal visual inspector.
- **Zero-Barrier Judge Experience:** Includes pre-loaded demo buttons for instant evaluation.

### ⚙️ Backend & AI Engine

- **Node.js Server (`server.js`):** Asynchronous HTTP server with zero runtime external dependencies for instant portability.
- **Gemini Multimodal Vision:** Unified evaluation pipeline inspecting copy, logo boundaries, and color hex drift.
- **Cryptographic Verification:** SHA-256 digest hashing for immutable compliance audit trails.

## 4 Explanation of Changes Made to Original Plan

During Round 1, our conceptual proposal focused primarily on static text highlighting and standard guideline lookup. In building the Round 2 MVP, we made three deliberate design enhancements to maximize real enterprise value:

- **From Passive Warnings to Consequence Simulation:** Traditional checkers output dry error codes. We built the *PR Blast Radius & Crisis Simulator* to visualize the concrete cost of failure (viral backlash and FTC regulatory exposure), giving non-technical executives an immediate business case.
- **From Rigid LLM Rewrites to Multi-Agent Negotiation:** Copywriters frequently reject AI-generated rewrites because they remove creative punch. We introduced the *Maverick vs. Guardian Debate* with a live compromise slider, allowing creators to balance legal safety with marketing conversion hooks.
- **From Simple Status Toggles to Cryptographic Passports:** Replaced standard "Approved" boolean flags with verifiable SHA-256 digital passports to prevent post-approval content tampering in enterprise publishing pipelines.

## 5 Three Major Improvements for Round 3 (The Mastermind Roadmap)

As mandated by the competition rules, these three major additions represent meaningful architectural capabilities:

### IMPROVEMENT 1 Self-Healing Playbook

**Auto-PR to Git & Figma:** When the system detects repeated team blindspots across 50+ posts, it autonomously drafts GitHub Pull Requests to update `brand-guidelines.md` and pushes design token patches directly into Figma.

### IMPROVEMENT 2 10-Persona Jury

**Synthetic Audience Simulation:** Expands the 2-agent debate into an autonomous panel of 10 synthetic AI personas (FTC auditor, skeptical enterprise buyer, competitor PR lead, tech critic) to predict virality, controversy, and sentiment before ad spend.

### IMPROVEMENT 3 ZK Brand Proofs

**Zero-Knowledge Publishing Mesh:** Issues cryptographic proofs allowing enterprises to verify compliance with external agencies and ad platforms without exposing confidential embargoed launch data, with direct automated publishing to CMS and social networks.

**BrandCircuit • Round 2 Build Submission** • Developed for the Inkloom AI Hackathon (October 2026)